

Appendix-1

Table S1. Exit-to-Pulse Survey Item Alignment, Theme Grouping, and Risk Coding

The table below can be used as **Supplementary Table S1**. It follows the 32-item structure used in the WYDOT prototype tool and keeps the same eight HR themes used in the manuscript.

Theme	Exit survey item	Pulse survey wording	Scale	Risk coding
Work Culture and Team	Cooperation within your department/program	Cooperation within my department/program is effective.	1–4	Lower = higher risk
Work Culture and Team	Cooperation with other departments/programs	Cooperation with other departments/programs is effective.	1–4	Lower = higher risk
Work Culture and Team	Members of my team worked well together	Members of my team work well together.	1–4	Lower = higher risk
Work Culture and Team	I was treated as a valuable member of WYDOT	I feel treated as a valuable member of WYDOT.	1–4	Lower = higher risk
Compensation and Career	Rate of pay for your job	I am satisfied with the rate of pay for my job.	1–4	Lower = higher risk
Compensation and Career	Career development/advancement opportunities	I have adequate career development and advancement opportunities.	1–4	Lower = higher risk
Compensation and Career	WYDOT's performance review system	WYDOT's performance review system supports fair evaluation and growth.	1–4	Lower = higher risk
Compensation and Career	Supervisor provided recognition/appreciated my work	My supervisor provides recognition and appreciation for my work.	1–4	Lower = higher risk
Supervisor/Leadership	Supervisor let me know when I was doing a good job	My immediate supervisor lets me know when I am doing a good job.	1–4	Lower = higher risk
Supervisor/Leadership	I felt free to suggest improvements to my supervisor	I feel free to suggest improvements to my supervisor.	1–4	Lower = higher risk
Supervisor/Leadership	I felt comfortable speaking with my immediate supervisor	I feel comfortable speaking with my immediate supervisor.	1–4	Lower = higher risk
Supervisor/Leadership	Employee problems and complaints were resolved promptly	Employee problems and complaints are resolved promptly in my department.	1–4	Lower = higher risk
Supervisor/Leadership	Supervisor was sensitive to employees' needs	My supervisor is sensitive to employees' needs.	1–4	Lower = higher risk
Supervisor/Leadership	Supervisor was receptive to open communication	My supervisor is receptive to open communication.	1–4	Lower = higher risk

Supervisor/Leadership	I felt comfortable speaking with upper management	I feel comfortable speaking with upper management.	1–4	Lower = higher risk
Supervisor/Leadership	Supervisor provided feedback on performance	My supervisor provides useful feedback on my performance.	1–4	Lower = higher risk
Training and Development	Personal job training provided	I receive adequate personal job training.	1–4	Lower = higher risk
Training and Development	Training programs	WYDOT provides effective training programs.	1–4	Lower = higher risk
Training and Development	I received proper training to perform my job effectively	I receive proper training to perform my job effectively.	1–4	Lower = higher risk
Working Conditions and Resources	Equipment, materials, resources, and facilities provided	I have adequate equipment, materials, resources, and facilities.	1–4	Lower = higher risk
Working Conditions and Resources	Physical working conditions	My physical working conditions are satisfactory.	1–4	Lower = higher risk
Working Conditions and Resources	Job duties and responsibilities were clearly defined	My job duties and responsibilities are clearly defined.	1–4	Lower = higher risk
Working Conditions and Resources	Ability to get resources needed to do my job	I am able to get the resources needed to do my job.	1–4	Lower = higher risk
Safety and Job Security	WYDOT supported a culture of safety	WYDOT supports a culture of safety.	1–4	Lower = higher risk
Safety and Job Security	Supervisor provided an environment of safety	My supervisor provides an environment of safety.	1–4	Lower = higher risk
Safety and Job Security	WYDOT provided me with job security	I feel that WYDOT provides me with job security.	1–4	Lower = higher risk
Communication and Information	Kept informed about department policies and procedures	I am kept well informed about department policies and procedures.	1–4	Lower = higher risk
Communication and Information	Supervisor followed WYDOT policies	My supervisor follows WYDOT policies.	1–4	Lower = higher risk
Communication and Information	Comfortable speaking with Human Resources	I feel comfortable speaking with Human Resources.	1–4	Lower = higher risk
HR/Fairness/Problem Resolution	Supervisor demonstrated fair and equal treatment	My supervisor demonstrates fair and equal treatment.	1–4	Lower = higher risk
HR/Fairness/Problem Resolution	Supervisor developed cooperation and teamwork	My supervisor develops cooperation and teamwork.	1–4	Lower = higher risk
HR/Fairness/Problem Resolution	Complaints and problems were resolved effectively	Complaints and problems are resolved effectively.	1–4	Lower = higher risk

TABLE S2 Theme-to-Strategy Recommendation Logic

High-Risk HR Theme	Example Problem Indicator	Recommended Retention Strategy
Compensation and Career	Employees cite low pay and no clear promotion path	Conduct pay benchmarking against peer DOTs; establish individual development plans; create transparent promotion criteria
Supervisor / Leadership	Exit surveys note a lack of supervisor support and poor communication from management	Implement supervisory coaching programs; conduct 360-degree leadership reviews; train managers in constructive feedback
Training and Development	Staff report no onboarding support and limited skill development access	Launch structured onboarding tracks; provide tuition reimbursement; establish internal mentorship networks
Work Culture and Team	Employees describe disconnected teams and lack of organizational pride	Invest in team-building initiatives; recognize contributions publicly; strengthen diversity and inclusion programs
Working Conditions and Resources	Field employees report inadequate equipment and excessive workload	Conduct equipment audits; introduce workload-monitoring protocols; offer schedule flexibility where feasible
Safety and Job Security	Respondents fear job elimination or express physical safety concerns	Communicate workforce stability commitments; update field safety standards; establish anonymous hazard reporting
Communication and Information	Employees unaware of organizational changes or policy updates	Launch regular town halls; distribute policy digests; create two-way feedback mechanisms
HR / Fairness / Problem Resolution	Staff perceive HR as unresponsive and processes as unfair	Audit grievance-resolution timelines; publish HR process transparency reports; appoint employee ombudsperson

Table S3. Prompt Architecture for the LLM-Based HR Assistant

Prompt block	Input supplied to the LLM	Purpose in the HR decision-support workflow	Example content used in this study
System role and task boundary	Fixed instruction defining the assistant as a WYDOT HR decision-support assistant	Establishes that the LLM explains model outputs but does not make independent employment decisions	“You are an HR decision-support assistant. Explain turnover-risk results using only the supplied model output, theme definitions, and strategy database.”
User query	Natural-language question entered by the HR practitioner	Identifies the specific information need	“Why is this employee high risk?”; “What should HR do for compensation-related risk?”
Model-output block	Turnover risk probability, risk category, prediction threshold, and classification context	Grounds the response in the predictive model rather than free-form reasoning	Risk probability = 0.72; Risk category = High; Operational categories: Low < 0.35, Medium 0.35–0.65, High ≥ 0.65
Theme-diagnostic block	Eight HR theme scores and top-ranked risk themes	Allows the LLM to explain which organizational domains are driving the risk	Compensation and Career = highest risk; HR/Fairness/Problem Resolution = second-highest risk
Survey-item explanation block	Lowest-rated survey items or strongest coefficient-based contributors	Connects the theme-level explanation to specific employee survey responses	Low scores on rate of pay, career development opportunities, and problem-resolution items
Strategy-retrieval block	Relevant strategies from the 102-strategy database, including HR theme, adoption count, effectiveness rating, scope, and resource intensity	Ensures recommendations are selected from peer DOT and literature-supported strategies	Pay benchmarking, transparent promotion pathways, individual development plans, HR process transparency
Safety and responsible-use block	Restrictions on what the LLM is not allowed to infer or recommend	Prevents unsupported, punitive, or discriminatory outputs	Do not infer protected characteristics; do not recommend termination or disciplinary action; state that all recommendations require HR review

Output-format block	Required response structure	Produces consistent, readable explanations for non-technical HR users	1) risk summary, 2) main drivers, 3) recommended actions, 4) caution/HR review note
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Appendix-2

AnyLogic Hierarchical Synthetic Stayer Pulse-Survey Generation Logic

For the workforce retention decision-support manuscript

S1. Purpose of the Supplementary Logic

This supplementary file documents the revised AnyLogic logic used to generate hypothetical current-employee pulse-survey profiles for the proof-of-concept leaver-stayer modeling dataset. The revised generator is hierarchical rather than purely item-independent: each simulated employee first receives an overall workplace-perception tendency, then theme-level tendencies, and finally item-level responses conditional on the relevant theme. **This design was introduced to reduce artificial independence among simulated survey items and to produce more realistic within-profile response structure.**

Important interpretation: The simulated stayer profiles are hypothetical comparison profiles and do not constitute observed current-employee outcomes. The generated data support prototype development and robustness testing only. Model outputs based on these profiles should be interpreted as leaver-synthetic-stayer separability, not as validated operational turnover prediction.

S2. Revised Synthetic-Stayer Generation Design

p	Step	Component	Logic
1	Overall tendency	Overall tendency	Assign each simulated employee one overall workplace-perception score on a 1-4 scale. This creates an employee-level response tendency.
2	Theme-level tendency	Theme-level tendency	Generate eight HR-theme scores conditional on the overall score. Theme scores can be one point lower, equal to, or one point higher than the overall score.
3	Item-level response	Item-level response	Generate each survey item from its corresponding theme score. Most item responses stay equal or adjacent to the theme score, with a small random-noise probability.
4	Text conversion	Text conversion	Convert numeric values to the original response labels, such as Poor/Fair/Good/Excellent or Strongly disagree/Disagree/Agree/Strongly agree.
5	CSV export	CSV export	Export 300 synthetic stayer profiles with Employee_ID and 32 aligned survey-item columns.

The generator was not revised to force any specific HR theme to become the strongest model result. The objective was to reduce unrealistic item independence and improve the methodological defensibility of the synthetic comparison group.

S3. Theme-Level Probability Settings

For each theme, themeFromOverall(overall, pDown, pSame, pUp) generates the theme score relative to the employee-level overall tendency. The probabilities below define whether the theme score moves one point down, stays the same, or moves one point up.

Element/theme	Probability setting	Reason
Overall workplace perception	sample4(0.04, 0.16, 0.50, 0.30)	Initial employee-level tendency
Compensation and Career	pDown = 0.30, pSame = 0.55, pUp = 0.15	Slightly weaker among stayers because pay/career concerns can remain in public agencies
Supervisor/Leadership	pDown = 0.15, pSame = 0.65, pUp = 0.20	Generally favorable but still variable
Training and Development	pDown = 0.20, pSame = 0.60, pUp = 0.20	Balanced theme variation
Work Culture and Team	pDown = 0.15, pSame = 0.65, pUp = 0.20	Generally favorable but still variable
Working Conditions and Resources	pDown = 0.18, pSame = 0.62, pUp = 0.20	Moderate favorable shift
Safety and Job Security	pDown = 0.22, pSame = 0.63, pUp = 0.15	Not made unrealistically perfect; allows safety/job-security concerns among stayers
Communication and Information	pDown = 0.18, pSame = 0.62, pUp = 0.20	Moderate favorable shift
HR/Fairness/Problem Resolution	pDown = 0.20, pSame = 0.60, pUp = 0.20	Balanced theme variation

S4. Generator-Side Theme-to-Item Mapping

The following mapping was used in the revised generator. Item names correspond to the exported CSV column names.

HR theme	No . of items	Mapped survey items
Work Culture and Team	5	Cooperation_within_department; Cooperation_with_other_departments; Treated_as_valuable_member; Free_to_suggest_changes; Team_worked_well_together
Training and Development	3	Personal_job_training; Training_programs; Proper_training
Working Conditions and Resources	4	Equipment_provided; Physical_working_conditions; Ease_of_getting_resources; Job_duties_clearly_defined
Compensation and Career	3	Performance_review_system; Rate_of_pay; Career_development_opportunities
HR/Fairness/Problem Resolution	2	Problems_complaints_resolved; ComfortableSpeaking_with_HR
Communication and Information	2	ComfortableSpeaking_with_upper_management; Kept_well_informed
Safety and Job Security	3	Job_security; Safety_culture; Supervisor_environment_safety

Supervisor/ Leadership	10	Supervisor_let_me_know_good_job; Supervisor_provided_recognition; Supervisor_sensitive_to_needs; Supervisor_open_communication; Supervisor_fair_equal_treatment; Supervisor_developed_teamwork	Comfortable_speaking_with_supervisor; Supervisor_resolved_complaints; Supervisor_feedback_performance; Supervisor_followed_policies;
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S5. Response-Scale Text Mapping

Item type	Numeric-to-text mapping
Rating items	1 = Poor; 2 = Fair; 3 = Good; 4 = Excellent
Agreement items	1 = Strongly disagree; 2 = Disagree; 3 = Agree; 4 = Strongly agree
Ease item	1 = Very difficult; 2 = Difficult; 3 = Easy; 4 = Very easy
Supervisor-frequency items	1 = Almost never; 2 = Sometimes; 3 = Often; 4 = Almost always

S6. Generated Output Format Check

Check	Result
Number of synthetic profiles	300
Total columns	33
Survey-item columns	32
Employee_ID range	1-300
Invalid response-text values	0
Overall mean item score	2.949 / 4
Row-mean range	1.312 to 3.812

Note: These checks confirm file structure, response-text validity, and simulated response distribution only. They do not validate turnover prediction accuracy.

Supplementary Table S1. Theme means and average within-theme correlations in the revised generated dataset.

Theme	No. of items	Mean score	Average correlation	within-theme
Work Culture and Team	5	3.006	0.621	
Training and Development	3	2.963	0.620	
Working Conditions and Resources	4	2.932	0.608	
Compensation and Career	3	2.829	0.671	
HR/Fairness/Problem Resolution	2	2.937	0.650	
Communication and Information	2	2.930	0.513	
Safety and Job Security	3	2.884	0.664	
Supervisor/Leadership	10	2.984	0.626	

S7. AnyLogic Helper Functions

Implementation note: In AnyLogic, each function below should be created as a Function object inside Main. Only the function body should be pasted into the Action/body field; do not paste the Java method signature into the body.

sample4(p1, p2, p3, p4)

```
// Return type: int
// Arguments: double p1, double p2, double p3, double p4
// Purpose: Draw a four-point response from user-specified probabilities.
```

```
double u = uniform(0, 1);
```

```
if (u < p1) return 1;
else if (u < p1 + p2) return 2;
else if (u < p1 + p2 + p3) return 3;
else return 4;
```

sample5(p1, p2, p3, p4, p5) - optional, not used in final four-point export

```
// Return type: int
// Arguments: double p1, double p2, double p3, double p4, double p5
// Purpose: Optional five-point sampler. Retained only if needed for future sensitivity tests.
```

```
double u = uniform(0, 1);
```

```
if (u < p1) return 1;
else if (u < p1 + p2) return 2;
else if (u < p1 + p2 + p3) return 3;
else if (u < p1 + p2 + p3 + p4) return 4;
else return 5;
```

ratingText(score)

```
// Return type: String
// Arguments: int score
```

```
if (score == 1) return "Poor";
else if (score == 2) return "Fair";
else if (score == 3) return "Good";
else return "Excellent";
```


agreementText(score)

```
// Return type: String
// Arguments: int score

if (score == 1) return "Strongly disagree";
else if (score == 2) return "Disagree";
else if (score == 3) return "Agree";
else return "Strongly agree";
```

easeText(score)

```
// Return type: String
// Arguments: int score

if (score == 1) return "Very difficult";
else if (score == 2) return "Difficult";
else if (score == 3) return "Easy";
else return "Very easy";
```

supervisorText(score)

```
// Return type: String
// Arguments: int score

if (score == 1) return "Almost never";
else if (score == 2) return "Sometimes";
else if (score == 3) return "Often";
else return "Almost always";
```

clamp4(x)

```
// Return type: int
// Arguments: int x
// Purpose: Keep generated values inside the 1-4 scale.

if (x < 1) return 1;
if (x > 4) return 4;
return x;
```

themeFromOverall(overall, pDown, pSame, pUp)

```
// Return type: int
// Arguments: int overall, double pDown, double pSame, double pUp
// Purpose: Generate a theme-level tendency conditional on the employee-level tendency.
```

```
double u = uniform(0, 1);

if (u < pDown) {
    return clamp4(overall - 1);
} else if (u < pDown + pSame) {
    return clamp4(overall);
} else {
    return clamp4(overall + 1);
}
```

itemFromTheme(themeScore)

```
// Return type: int
// Arguments: int themeScore
// Purpose: Generate an item-level response conditional on the theme-level tendency.
// Logic: 70% same as the theme score, 25% adjacent value, 5% general random variation.
```

```
double u = uniform(0, 1);

if (u < 0.70) {
    return themeScore;
} else if (u < 0.95) {

    if (themeScore == 1) return 2;
    if (themeScore == 4) return 3;

    if (uniform(0, 1) < 0.5) {
        return themeScore - 1;
    } else {
        return themeScore + 1;
    }

} else {
    return sample4(0.06, 0.18, 0.50, 0.26);
}
```

S8. Event Action Code for CSV Generation

Event name used in the model: generateItemBasedPulseSurvey. The output filename is intentionally changed to hierarchical_revised to distinguish it from the earlier item-independent dataset.

```
try {

    // Save file to Documents folder
    java.io.File outDir = new java.io.File(
        System.getProperty("user.home") + java.io.File.separator + "Documents"
```

```

);

if (!outDir.exists()) {
    outDir.mkdirs();
}

java.io.File file = new java.io.File(
    outDir,
    "anylogic_hierarchical_revised_hypothetical_pulse_survey.csv"
);

java.io.PrintWriter pw = new java.io.PrintWriter(file);

traceln("Generating hierarchical hypothetical stayer pulse survey data...");
traceln("File will be saved here: " + file.getAbsolutePath());

// Header row
pw.println(
    "Employee_ID," +
    "Cooperation_within_department," +
    "Cooperation_with_other_departments," +
    "Personal_job_training," +
    "Equipment_provided," +
    "Performance_review_system," +
    "Training_programs," +
    "Rate_of_pay," +
    "Career_development_opportunities," +
    "Physical_working_conditions," +
    "Ease_of_getting_resources," +
    "Treated_as_valuable_member," +
    "Supervisor_let_me_know_good_job," +
    "Free_to_suggest_changes," +
    "Proper_training," +
    "Problems_complaints_resolved," +
    "Comfortable_speaking_with_supervisor," +
    "Comfortable_speaking_with_upper_management," +
    "Comfortable_speaking_with_HR," +
    "Kept_well_informed," +
    "Job_security," +
    "Team_worked_well_together," +
    "Safety_culture," +
    "Job_duties_clearly_defined," +
    "Supervisor_provided_recognition," +
    "Supervisor_resolved_complaints," +
    "Supervisor_sensitive_to_needs," +
    "Supervisor_feedback_performance," +
    "Supervisor_open_communication," +
    "Supervisor_followed_policies," +
    "Supervisor_fair_equal_treatment," +
    "Supervisor_developed_teamwork," +
    "Supervisor_environment_safety"
);

int n = 300;

for (int id = 1; id <= n; id++) {

```

```

// Step 1: overall workplace perception tendency
// 1 = very unfavorable, 4 = very favorable
int overall = sample4(0.04, 0.16, 0.50, 0.30);

// Step 2: theme-level tendencies
// Compensation is intentionally a little weaker because pay/career concern
// may still exist among active public-sector employees.
// Safety is no longer made unrealistically perfect.
int compCareerTheme = themeFromOverall(overall, 0.30, 0.55, 0.15);
int supervisorTheme = themeFromOverall(overall, 0.15, 0.65, 0.20);
int trainingTheme = themeFromOverall(overall, 0.20, 0.60, 0.20);
int cultureTheme = themeFromOverall(overall, 0.15, 0.65, 0.20);
int workingCondTheme = themeFromOverall(overall, 0.18, 0.62, 0.20);
int safetyTheme = themeFromOverall(overall, 0.22, 0.63, 0.15);
int communicationTheme = themeFromOverall(overall, 0.18, 0.62, 0.20);
int hrFairnessTheme = themeFromOverall(overall, 0.20, 0.60, 0.20);

// Step 3: item-level responses conditional on theme tendency

// Work Culture and Team
int q1 = itemFromTheme(cultureTheme); // Cooperation within department
int q2 = itemFromTheme(cultureTheme); // Cooperation with other departments
int q11 = itemFromTheme(cultureTheme); // Treated as valuable member
int q13 = itemFromTheme(cultureTheme); // Free to suggest changes
int q21 = itemFromTheme(cultureTheme); // Team worked well together

// Training and Development
int q3 = itemFromTheme(trainingTheme); // Personal job training
int q6 = itemFromTheme(trainingTheme); // Training programs
int q14 = itemFromTheme(trainingTheme); // Proper training

// Working Conditions and Resources
int q4 = itemFromTheme(workingCondTheme); // Equipment provided
int q9 = itemFromTheme(workingCondTheme); // Physical working conditions
int q10 = itemFromTheme(workingCondTheme); // Ease of getting resources
int q23 = itemFromTheme(workingCondTheme); // Job duties clearly defined

// Compensation and Career
int q5 = itemFromTheme(compCareerTheme); // Performance review system
int q7 = itemFromTheme(compCareerTheme); // Rate of pay
int q8 = itemFromTheme(compCareerTheme); // Career development opportunities

// HR / Fairness / Problem Resolution
int q15 = itemFromTheme(hrFairnessTheme); // Problems/complaints resolved
int q18 = itemFromTheme(hrFairnessTheme); // Comfortable speaking with HR

// Communication and Information
int q17 = itemFromTheme(communicationTheme); // Comfortable speaking with upper management
int q19 = itemFromTheme(communicationTheme); // Kept well informed

// Safety and Job Security
int q20 = itemFromTheme(safetyTheme); // Job security
int q22 = itemFromTheme(safetyTheme); // Safety culture
int q32 = itemFromTheme(safetyTheme); // Supervisor environment safety

// Supervisor / Leadership

```

```

int q12 = itemFromTheme(supervisorTheme); // Supervisor let me know good job
int q16 = itemFromTheme(supervisorTheme); // Comfortable speaking with supervisor
int q24 = itemFromTheme(supervisorTheme); // Supervisor provided recognition
int q25 = itemFromTheme(supervisorTheme); // Supervisor resolved complaints
int q26 = itemFromTheme(supervisorTheme); // Supervisor sensitive to needs
int q27 = itemFromTheme(supervisorTheme); // Supervisor feedback performance
int q28 = itemFromTheme(supervisorTheme); // Supervisor open communication
int q29 = itemFromTheme(supervisorTheme); // Supervisor followed policies
int q30 = itemFromTheme(supervisorTheme); // Supervisor fair/equal treatment
int q31 = itemFromTheme(supervisorTheme); // Supervisor developed teamwork

// Write one hypothetical stayer employee response row
pw.println(
    id + "," +
    ratingText(q1) + "," +
    ratingText(q2) + "," +
    ratingText(q3) + "," +
    ratingText(q4) + "," +
    ratingText(q5) + "," +
    ratingText(q6) + "," +
    ratingText(q7) + "," +
    ratingText(q8) + "," +
    ratingText(q9) + "," +
    easeText(q10) + "," +
    agreementText(q11) + "," +
    agreementText(q12) + "," +
    agreementText(q13) + "," +
    agreementText(q14) + "," +
    agreementText(q15) + "," +
    agreementText(q16) + "," +
    agreementText(q17) + "," +
    agreementText(q18) + "," +
    agreementText(q19) + "," +
    agreementText(q20) + "," +
    agreementText(q21) + "," +
    agreementText(q22) + "," +
    agreementText(q23) + "," +
    supervisorText(q24) + "," +
    supervisorText(q25) + "," +
    supervisorText(q26) + "," +
    supervisorText(q27) + "," +
    supervisorText(q28) + "," +
    supervisorText(q29) + "," +
    supervisorText(q30) + "," +
    supervisorText(q31) + "," +
    supervisorText(q32)
);
}

pw.close();

traceln("Hierarchical hypothetical stayer pulse survey generated successfully.");
traceln("File saved here: " + file.getAbsolutePath());

getEngine().stop();

} catch (Exception ex) {

    traceln("Export error: " + ex.getMessage());

```

```

    getEngine().stop();
}

```

S9. Recommended Manuscript Description

The following paragraph can be used in the Methods or Supplementary Methods section:

The AnyLogic generator was revised from an item-independent probability sampler to a hierarchical profile generator. Each simulated current-employee profile was first assigned an overall workplace-perception tendency. Theme-level tendencies were then generated for the eight HR domains, and item-level responses were sampled conditionally on the corresponding theme tendency. This structure introduced within-profile and within-theme correlation among related survey items, reducing the artificial independence pattern that can arise when all items are generated separately from fixed marginal probabilities. The simulated stayer profiles were used only as a proof-of-concept comparison group and should not be interpreted as observed current-employee attitudes or validated future turnover outcomes.

S10. Reporting Guidance for Results

Category	Statement
Safe interpretation	The revised generator reduces artificial item independence by introducing employee-level and theme-level response dependence.
Safe interpretation	Compensation and Career can be reported as the leading self-reported exit-survey concern if supported by the leaver-only exit-survey data.
Safe interpretation	Any theme emerging from the leaver-synthetic-stayer model should be described as a model-derived discriminator, not as a confirmed causal turnover driver.
Avoid	Do not state that the synthetic-stayer model validates employee-level turnover probabilities.
Avoid	Do not state that the generator proves a particular theme is the true cause of leaving.